



PHD Research Problem Formulation Presentation

ADAPTIVE CONTEXT REASONING SYSTEM (ACRS)

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Proposal stage. No empirical performance claims. Live Indian Open Government Data is a methodological constraint, not a result.

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Attribute and Details as recorded for the FET research-proposal template (Gowrishankar PRP2 field names).

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Contents of Slides

Nine required sections plus the FET mapping To Solve the Research Gap. Literature Review follows the Evidence 1, Evidence 2, ... template. After the literature summary, each limitation is closed by a named ACRS module. Research Methodology is expanded for each objective (O1 SATR, O2 APRR, O3 MNCD, O4 FCNP), the integrated loop, the exact repository formulas, and two worked traces (ToolBench-schema ranking; live data.gov.in). Objective order is SATR → APRR → MNCD → FCNP (not SMART).

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Introduction

This proposal treats multi-agent LLM systems as a computer-science systems problem: not a new foundation model, but a missing orchestration layer between session memory, specialist routing, live tools, and context growth.

Wang et al. survey LLM-based autonomous agents as a construction problem in *Frontiers of Computer Science* (2024, doi:10.1007/s11704-024-40231-1). He, Treude and Lo review LLM-based multi-agent systems for software engineering in *ACM TOSEM* (2025, doi:10.1145/3712003). Guo et al. survey LLM multi-agent profiling and communication at *IJCAI-24* (doi:10.24963/ijcai.2024/890). Chang and Geng give transactional context management for multi-agent LLM planning in *PVLDB 2025* (doi:10.14778/3750601.3750611).

Wu et al. introduce AutoGen as a conversation-driven programming framework in which agents exchange messages until a stopping condition (*COLM 2024*). Hong et al. encode Standard Operating Procedures into MetaGPT so that a software-company metaphor produces structured artefacts (*ICLR 2024*). Qian et al. organise ChatDev as a chat-chain of organisational roles (*ACL 2024*). Li et al. study communicative agents in *CAMEL* (*NeurIPS 2023*).

Tool use is a parallel line. Qin et al. release ToolLLM / ToolBench: 16k+ REST APIs, a DFSDT planner, and ToolEval (*ICLR 2024*). Zheng et al. add ToolRerank over ToolLLM candidates (*LREC-COLING 2024*). Learned routers (RouteLLM, MasRouter, PILOT) pick models or collaboration modes. LLMingua shortens prompts by token importance (*EMNLP 2023*).

Those stacks still leave four operational surfaces underspecified as one contract: session-tool fusion, a training-free specialist posterior, fail-loud live Indian Open Government Data, and a mesh prune that writes a residue back into retrieval. The next slides record that literature as Evidence 1, Evidence 2, ... in the FET template. Every Evidence block is a journal article or a top international conference paper.

ACRS is proposed as that missing layer. The integration order is SATR → APRR → MNCD → FCNP. The live demonstration corpus is Agriculture on data.gov.in. This deck does not claim a leaderboard number.

Literature Review

FET template. Each paper is one Evidence block: Author(s), Year, Title, Publication, Objective, Methodology, Findings, Limitations, To solve the research gap. Publication is the journal or the full international conference proceedings title — not a preprint report.

Evidence 1

Author(s): Wang, L., Ma, C., Feng, X., Zhang, Z., Yang, H., Zhang, J., Chen, Z., Tang, J., Chen, X., Lin, Y., Zhao, W. X., Wei, Z., and Wen, J.

Year: 2024

Title: A survey on large language model based autonomous agents

Publication: Frontiers of Computer Science, 18, article 186345 (2024). doi:10.1007/s11704-024-40231-1

Objective: Unify how LLM-based autonomous agents are constructed, applied, and evaluated.

Methodology: Journal survey of agent construction: a profile–memory–planning–action framework covering social science, natural science, and engineering applications.

Findings: Most published agents share a construction template, but coordination remains an open systems problem rather than a solved runtime.

Limitations: The survey organises single- and multi-agent construction. It does not specify a closed SATR → APRR → MNCD → FCNP contract over live Indian Open Government Data.

To solve the research gap: Integrated ACRS — four named modules on one turn, with O3 MNCD citing a verified data.gov.in UUID.

Evidence 2

Author(s): He, J., Treude, C., and Lo, D.

Year: 2025

Title: LLM-Based Multi-Agent Systems for Software Engineering: Literature Review, Vision, and the Road Ahead

Publication: ACM Transactions on Software Engineering and Methodology, 34(5), May 2025. doi:10.1145/3712003

Objective: Map LLM-based multi-agent (LMA) systems across the software development lifecycle.

Methodology: Systematic review of primary LMA studies in software engineering, plus case studies of current frameworks and a two-phase research agenda.

Findings: Role-specialised LMA systems can divide software work, but agent synergy and trustworthiness remain research gaps.

Limitations: The environment is a software project, not a ministry API. Orchestration is reviewed as chat/SOP collaboration, not a live (toolld, score, citation).

To solve the research gap: O3 MNCD — the vote object is a live tool identifier on data.gov.in, not a software-engineering role in a chat chain.

Literature Review — Evidence 3 & Evidence 4

Continuation. Evidence 3 & Evidence 4 in the same Evidence template.

Evidence 3

Author(s): Guo, T., Chen, X., Wang, Y., Chang, R., Pei, S., Chawla, N. V., Wiest, O., and Zhang, X.

Year: 2024

Title: Large Language Model based Multi-Agents: A Survey of Progress and Challenges

Publication: Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI-24), Survey Track, pages 8048–8057.
doi:10.24963/ijcai.2024/890

Objective: Survey how LLM-based multi-agent systems are profiled, how they communicate, and how their capacities grow.

Methodology: IJCAI survey track: domains and environments, agent profiling, communication mechanisms, and capacity-growth methods.

Findings: Published MAS work clusters on conversation, role profiles, and simulated worlds; live tool-identifier consensus is not the unit of analysis.

Limitations: Communication is reviewed as natural-language exchange. There is no fail-loud live Open Government Data GET as the coordination object.

To solve the research gap: O3 MNCD — gossip (toolId, score), score-sum consensus, then a fail-loud live data.gov.in GET.

Evidence 4

Author(s): Chang, E. Y., and Geng, L.

Year: 2025

Title: SagaLLM: Context Management, Validation, and Transaction Guarantees for Multi-Agent LLM Planning

Publication: Proceedings of the VLDB Endowment (PVLDB), 18(12):4874–4886, 2025. doi:10.14778/3750601.3750611

Objective: Give multi-agent LLM planners persistent context, validation, and compensable transactions.

Methodology: Saga-style checkpointing and compensation around multi-LLM planning, with independent validators and state tracking.

Findings: Transactional context management can recover multi-agent plans from disruption better than unconstrained chat loops.

Limitations: Orchestration is a planning/transaction runtime. It does not maintain a training-free specialist posterior over Indian OGD tool families, nor cite a ministry UUID.

To solve the research gap: O2 APRR plus O3 MNCD — Dirichlet–Thompson specialist hops, then a verified Agriculture UUID rather than a compensable software saga.

09/07/2026

Literature Review — Evidence 5 & Evidence 6

Continuation. Evidence 5 & Evidence 6 in the same Evidence template.

Evidence 5

Author(s): Wu, Q., Bansal, G., Zhang, J., Wu, Y., Li, B., Zhu, E., Jiang, L., Zhang, X., Zhang, S., Liu, J., Awadallah, A. H., White, R. W., Burger, D., and Wang, C.

Year: 2024

Title: AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversations

Publication: Proceedings of the First Conference on Language Modeling (COLM 2024)

Objective: Program LLM applications by composing multiple conversable agents.

Methodology: Agents exchange natural-language messages until a stopping condition; modes mix LLMs, humans, and tools.

Findings: Conversation is a working programming model for multi-agent LLM applications.

Limitations: The unit of coordination is a chat message, not a live (toolId, score, citation) on Indian Open Government Data.

To solve the research gap: O3 MNCD — gossip (toolId, score), score-sum consensus, then a fail-loud live data.gov.in GET.

Evidence 6

Author(s): Hong, S., Zhuge, M., Chen, J., Zheng, X., Cheng, Y., Wang, J., Zhang, C., Wang, Z., Yau, S. K. S., Lin, Z., Zhou, L., Ran, C., Xiao, L., Wu, C., and Schmidhuber, J.

Year: 2024

Title: MetaGPT: Meta Programming for A Multi-Agent Collaborative Framework

Publication: Proceedings of the Twelfth International Conference on Learning Representations (ICLR 2024)

Objective: Reduce role drift in multi-agent software workflows by encoding human SOPs.

Methodology: Standard Operating Procedures are written into prompt sequences; an assembly-line assigns roles.

Findings: Authored SOPs produce more coherent software artefacts than unconstrained chat agents.

Limitations: Who speaks next is designed in advance. The graph is not updated from session-local affinity after a live tool call.

To solve the research gap: O2 APRR — training-free specialist posterior updated after live outcomes, not an authored SOP graph.

Literature Review — Evidence 7 & Evidence 8

Continuation. Evidence 7 & Evidence 8 in the same Evidence template.

Evidence 7

Author(s): Qian, C., Liu, W., Liu, H., Chen, N., Dang, Y., Li, J., Yang, C., Chen, W., Su, Y., Cong, X., Xu, J., Li, D., Liu, Z., and Sun, M.

Year: 2024

Title: ChatDev: Communicative Agents for Software Development

Publication: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 15174–15186.
doi:10.18653/v1/2024.acl-long.810

Objective: Organise specialised LLM agents as a chat-chain for software design, coding, and testing.

Methodology: Chat chain (what to communicate) plus communicative dehallucination (how to communicate).

Findings: Natural-language design talk and code-level debug talk can be unified in one multi-agent loop.

Limitations: The environment is a codebase, not a ministry API with a fail-loud live GET.

To solve the research gap: O3 MNCD — the environment is a verified ministry UUID; empty filters fail loud instead of inventing rows.

Evidence 8

Author(s): Li, G., Hammoud, H. A. A. K., Itani, H., Khizbullin, D., and Ghanem, B.

Year: 2023

Title: CAMEL: Communicative Agents for “Mind” Exploration of Large Language Model Society

Publication: Advances in Neural Information Processing Systems 36 (NeurIPS 2023)

Objective: Enable autonomous cooperation among communicative agents with minimal human steering.

Methodology: Role-playing with inception prompting to keep agents on a human-specified task.

Findings: Inception prompting yields scalable multi-agent conversational data and cooperative behaviour.

Limitations: No first-class vote over tool identifiers backed by Open Government Data.

To solve the research gap: O3 MNCD — the vote object is a tool identifier backed by Open Government Data, not a role-play utterance.

Literature Review — Evidence 9 & Evidence 10

Continuation. Evidence 9 & Evidence 10 in the same Evidence template.

Evidence 9

Author(s): Qin, Y., Liang, S., Ye, Y., Zhu, K., Yan, L., Lu, Y., Lin, Y., Cong, X., Tang, X., Qian, B., Zhao, S., Hong, L., Tian, R., Xie, R., Zhou, J., Gerstein, M., Li, D., Liu, Z., and Sun, M.

Year: 2024

Title: ToolLLM: Facilitating Large Language Models to Master 16000+ Real-world APIs

Publication: Proceedings of the Twelfth International Conference on Learning Representations (ICLR 2024)

Objective: Give open LLMs general tool-use over a large real-world API catalogue.

Methodology: ToolBench: 16,464 RapidAPI REST endpoints; SBERT retriever; DFSDT planner; ToolEval protocol.

Findings: A public ranking library and planner exist for large-scale tool use.

Limitations: Retrieval is turn-amnesic. RapidAPI keys are not redistributable, so this lab cannot execute ToolBench endpoints live.

To solve the research gap: O1 SATR uses ToolBench as a ranking library only. O3 MNCD executes live Indian OGD, never RapidAPI replay.

Evidence 10

Author(s): Zheng, Y., Li, P., Liu, W., Liu, Y., Luan, J., and Wang, B.

Year: 2024

Title: ToolRerank: Adaptive and Hierarchy-Aware Reranking for Tool Retrieval

Publication: Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024), pages 16263–16273. ACL Anthology 2024.lrec-main.1413

Objective: Refine ToolLLM-style retrieval for seen versus unseen APIs and for tool-library hierarchy.

Methodology: Adaptive truncation of seen/unseen APIs plus hierarchy-aware concentration or diversity.

Findings: Reranking the SBERT shortlist improves downstream tool execution quality.

Limitations: Still query-only. Session co-activation and later write-back from a pruned mesh are unused.

To solve the research gap: O1 SATR — fuse co-activation after ToolRerank-style truncation. O4 FCNP writes the residue back.

Literature Review — Evidence 11 & Evidence 12

Continuation. Evidence 11 & Evidence 12 in the same Evidence template.

Evidence 11

Author(s): Ong, I., Almahairi, A., Wu, V., Chiang, W.-L., Wu, T., Gonzalez, J. E., Kadous, M. W., and Stoica, I.

Year: 2025

Title: RouteLLM: Learning to Route LLMs from Preference Data

Publication: Proceedings of the Thirteenth International Conference on Learning Representations (ICLR 2025)

Objective: Route each query between a stronger and a weaker LLM from human preference data.

Methodology: Trained router plus data augmentation; deployed as a frozen policy at inference.

Findings: Preference-trained routers can cut cost while holding response quality on public benchmarks.

Limitations: The object of routing is an LLM SKU, not a named Indian-OGD specialist with a training-free posterior.

To solve the research gap: O2 APRR — route named mandi / crop / rainfall specialists, not foundation-model SKUs.

Evidence 12

Author(s): Yue, Y., Zhang, G., Liu, B., Wan, G., Wang, K., Cheng, D., and Qi, Y.

Year: 2025

Title: MasRouter: Learning to Route LLMs for Multi-Agent Systems

Publication: Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 15549–15572.

doi:10.18653/v1/2025.acl-long.757

Objective: Unify collaboration mode, role allocation, and LLM choice as one MAS routing problem.

Methodology: Cascaded neural controller trained over multi-agent topologies.

Findings: A learned controller can assemble a cheaper MAS than a static multi-agent template.

Limitations: Requires training. It does not maintain a Dirichlet–Thompson matrix over mandi / crop / rainfall specialists.

To solve the research gap: O2 APRR — Dirichlet–Thompson posterior over named specialists; learned routers stay a future bake-off.

Literature Review — Evidence 13 & Evidence 14

Continuation. Evidence 13 & Evidence 14 in the same Evidence template.

Evidence 13

Author(s): Panda, P., Magazine, R., Devaguptapu, C., Takemori, S., and Sharma, V.

Year: 2025

Title: Adaptive LLM Routing under Budget Constraints

Publication: Findings of the Association for Computational Linguistics: EMNLP 2025, pages 23934–23949. doi:10.18653/v1/2025.findings-emnlp.1301.

Method name: PILOT (Preference-prior Informed LinUCB).

Objective: Treat LLM routing as a contextual bandit under a user budget.

Methodology: Preference-prior LinUCB in a shared query-LLM embedding space, plus an online cost policy.

Findings: Bandit routing can adapt without exhaustive inference of every LLM on every query.

Limitations: Routes foundation-model SKUs under a dollar budget, not tool-specialist agents after a live data.gov.in GET.

To solve the research gap: O2 APRR — specialist hops after SATR, updated from live MNCD outcomes, not SKU bandits under a dollar budget.

Evidence 14

Author(s): Jiang, H., Wu, Q., Lin, C.-Y., Yang, Y., and Qiu, L.

Year: 2023

Title: LLMingua: Compressing Prompts for Accelerated Inference of Large Language Models

Publication: Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, pages 13358–13376.

doi:10.18653/v1/2023.emnlp-main.825

Objective: Shorten long prompts while keeping task performance.

Methodology: Budget controller plus token-level iterative compression aligned to the target LLM.

Findings: Token-importance compression can reduce prompt length with limited quality loss.

Limitations: Deletes tokens before the LLM. It does not prune a mesh by conductance or write live citations back into retrieval.

To solve the research gap: O4 FCNP — prune the mesh by conductance and pin live citations into SATR, not token deletion.

Literature Review — Evidence 15 & Evidence 16

Continuation. Evidence 15 & Evidence 16 in the same Evidence template.

Evidence 15

Author(s): Park, J. S., O'Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., and Bernstein, M. S.

Year: 2023

Title: Generative Agents: Interactive Simulacra of Human Behavior

Publication: Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST 2023), Best Paper.
doi:10.1145/3586183.3606763

Objective: Give LLM agents a long-term memory stream for believable behaviour in a sandbox.

Methodology: Natural-language memory, reflection, and retrieval (recency, relevance, importance) inside a Sims-like town.

Findings: A memory stream supports individual plans and emergent social coordination among agents.

Limitations: Memory lives in a sandbox simulation. It is not written back into a tool retriever over ministry APIs.

To solve the research gap: O4 FCNP — pin live ministry citations into the next SATR prior, not a sandbox memory stream.

Evidence 16

Author(s): Tero, A., Takagi, S., Saigusa, T., Ito, K., Bebber, D. P., Fricker, M. D., Yumiki, K., Kobayashi, R., and Nakagaki, T.

Year: 2010

Title: Rules for Biologically Inspired Adaptive Network Design

Publication: Science 327(5964):439–442. doi:10.1126/science.1177894

Objective: Capture how Physarum polycephalum grows efficient, fault-tolerant transport networks.

Methodology: Feedback between tube conductance and protoplasmic flux; compared with the Tokyo rail system.

Findings: A local conductance update can yield globally efficient adaptive networks without a central planner.

Limitations: A biological transport model, not an LLM context pruner. ACRS uses it as a design heuristic, not as a claim that the mesh is an organism.

To solve the research gap: O4 FCNP — Physarum-inspired conductance as a design heuristic, with write-back into SATR.

Literature Review — Evidence 17 & Evidence 18

Continuation. Evidence 17 & Evidence 18 in the same Evidence template.

Evidence 17

Author(s): Guo, H., Woodruff, A., and Yadav, A.

Year: 2020

Title: Improving Lives of Indebted Farmers Using Deep Learning: Predicting Agricultural Produce Prices Using Convolutional Neural Networks (PECAD)

Publication: Proceedings of the AAAI Conference on Artificial Intelligence, 34(08):13294–13299. doi:10.1609/aaai.v34i08.7039

Objective: Use AGMARKNET price series as decision-support input for indebted farmers.

Methodology: Convolutional networks over scraped agricultural produce prices.

Findings: AGMARKNET is a real Indian decision-support corpus, not a toy table.

Limitations: A crop-price CNN is not a multi-agent live-OGD loop. Cited as domain precedent, not as a baseline to beat.

To solve the research gap: O3 MNCD — live AGMARKNET UUID as a citation. PECAD is domain precedent, not a baseline to beat.

Evidence 18

Author(s): Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., and Cao, Y.

Year: 2023

Title: ReAct: Synergizing Reasoning and Acting in Language Models

Publication: Proceedings of the Eleventh International Conference on Learning Representations (ICLR 2023)

Objective: Interleave reasoning traces with actions so a language model can use tools.

Methodology: Thought–action–observation cycles on the already-chosen tool set.

Findings: Reasoning-and-acting beats reason-only or act-only prompting on several agent tasks.

Limitations: Assumes the tool set is already determined. It does not fuse session co-activation into the next rank, nor cite a live data.gov.in UUID.

To solve the research gap: O1 SATR — choose the tool set from session memory before any ReAct-style acting. O3 MNCD cites the UUID.

Summary of Literature Review

Qualitative map only. This table names the gap each family leaves for ACRS; it does not claim a percentage improvement over any baseline. The next slide is the FET section To Solve the Research Gap.

Literature family	What it already does	What ACRS still has to add
MAS frameworks (AutoGen, MetaGPT, ChatDev, CAMEL)	Conversation, SOPs, software roles, inception prompting	A closed session–route–mesh–prune contract, not another chat runtime
MAS surveys and orchestration (Wang et al. FCS 2024; He et al. TOSEM 2025; Guo et al. IJCAI-24; SagaLLM PVLDB 2025)	Journal and flagship-conference accounts of how LLM agents are built, profiled, and given transactional context	A live Indian-OGD loop whose vote object is a tool identifier, not a software SDLC or a saga log
Tool learning (ToolLLM / ToolBench / ToolRerank)	Large-scale tool-use corpus and ToolEval-style ranking protocol	Session-fused ranking plus live Indian OGD execution (not RapidAPI replay)
Learned routers (RouteLLM, MasRouter, PILOT)	Train or bandit-route over model SKUs or collaboration modes	Training-free Dirichlet–Thompson posterior over named specialists
Prompt compression (LLMLingua)	Shorten tokens before the LLM	Prune a mesh by conductance and write the residue back into SATR
Indian agriculture DSS (PECAD)	Shows AGMARKNET as a real decision-support input	Live, fail-loud OGD inside a multi-agent loop — not a crop-yield model to beat

To Solve the Research Gap

To solve the research gaps identified in Evidence 1–18, the following work is proposed. Closing a gap is a named module on a closed loop, not a promised leaderboard number.

Research gap	Left open by	To solve the research gap
G1 — Turn-amnesic retrieval	Evidence 9, 10, 18	O1 SATR. Fuse session co-activation with semantic rank; write FCNP residue into the next prior.
G2 — Model / SOP routers	Evidence 6, 11, 12, 13	O2 APRR. Training-free Dirichlet–Thompson posterior over named specialists, not LLM SKUs.
G3 — Chat / star coordination	Evidence 1–5, 7, 8	O3 MNCD. Score-sum over live tool identifiers, then a fail-loud data.gov.in GET.
G4 — Token compression, no write-back	Evidence 14, 15, 16	O4 FCNP. Conductance prune of the mesh; pin live citations back into SATR.
G5 — No live Indian OGD loop	Evidence 9, 17	Integrated ACRS on verified Agriculture UUIDs. ToolBench is ranking-only; prices are never invented.

Integration order: SATR → APRR → MNCD → FCNP on one user turn. No NDCG, latency, or accuracy target is attached to any row.

Identified Research Problem

Five architectural gaps. Four named objectives. One integration contract. The previous slide maps each gap onto the module that closes it. Closing a gap is evidenced by a runnable loop and citable equations, not by a promised leaderboard number.

G1 — Turn-amnesic retrieval. ToolLLM SBERT (Qin et al., ICLR 2024) and ToolRerank (Zheng et al., LREC-COLING 2024) score each query independently. Session co-activation is unused. → Objective 1 SATR.

G2 — Routers pick models or authored SOPs, not tool-specialist agents with a training-free posterior. RouteLLM / PILOT / MasRouter / MetaGPT. → Objective 2 APRR.

G3 — MAS communication is star, SOP, or chat. No mesh vote whose object is a live tool identifier. AutoGen, MetaGPT, ChatDev, CAMEL. → Objective 3 MNCD.

G4 — Prompt compressors (LLMLingua, EMNLP 2023) do not write live citations back into retrieval. → Objective 4 FCNP.

G5 — No closed four-stage contract executed on journal-publishable Indian OGD (data.gov.in / AGMARKNET / IMD / DES), with ToolBench used only as a ranking library. → Integrated ACRS.

Gaps are architectural. This deck does not convert them into NDCG, latency, or accuracy targets.

Research Title & Aim

Adaptive Context Reasoning System (ACRS): A Structural Orchestration Layer for Multi-Agent LLM Ecosystems

Aim. Design, implement, and critically evaluate ACRS — a structural orchestration layer in which session-aware tool retrieval (SATR), training-free specialist routing (APRR), mesh consensus over tool identifiers (MNCD), and flow-coupled context pruning with write-back (FCNP) form a closed loop on live Indian Open Government Data. The proposal-stage claim is architectural completeness and live-pipeline integrity, not a leaderboard number.

Scope of the title. Adaptive = session-conditioned ranking and routing. Context = fused session memory plus mesh residue. Reasoning = specialist posterior plus score-sum consensus. System = four named modules on one contract.

What the title is not. It is not a new LLM, not a crop-yield model, and not a claim that ACRS already outperforms ToolLLM, MasRouter, or LLMLingua on a published leaderboard.

Domain lock for the live loop. Agriculture on data.gov.in (AGMARKNET and related verified UUIDs). Other sectors stay out of the inference path in this proposal.

Integration order that the aim implies. SATR (memory) → APRR (who acts) → MNCD (live evidence) → FCNP (prune and write back).

Research Objectives

To solve the research gap, four named modules are proposed. Each row is a thesis-sized design claim. Metric targets are deferred until a protocol is frozen.

ID	Objective	What will be designed
O1	SessionRerank+ (SATR)	The retrieval chapter of the thesis: given query q and session history H , return a ranked list of ToolBench-schema tools fused with a co-activation cache. The unit of publication is the fusion rule, not an NDCG target.
O2	Adaptive Probabilistic Routing Reinforcement (APRR)	The routing chapter: a training-free posterior over tool-specialist agents, updated from SATR scores and from live MNCD observations. The unit of publication is the Bayesian controller, not a Pareto chart.
O3	Mesh Network Context Diffusion (MNCD)	The consensus-and-execution chapter: gossip (toolId, score), score-sum consensus, live data.gov.in GET on verified UUIDs. The unit of publication is the protocol plus the citation contract.
O4	Flow-Coupled Network Pruning (FCNP)	The memory chapter: Kirchhoff/Physarum conductances prune the mesh trace and write surviving live citations back into SATR. The unit of publication is the coupling, not a token-percentage.

Objective 1 — SATR

The retrieval chapter of the thesis: given query q and session history H , return a ranked list of ToolBench-schema tools fused with a co-activation cache. The unit of publication is the fusion rule, not an NDCG target.

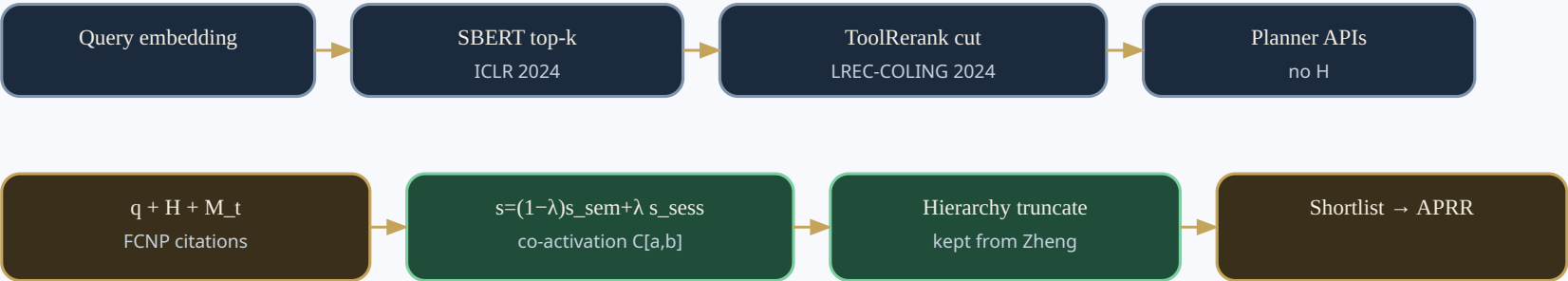
SOTA. Qin et al., ToolLLM / ToolBench (ICLR 2024): Sentence-BERT API retriever over 16,464 RapidAPI tools, then ToolLLaMA + DFSDT. Zheng et al., ToolRerank (LREC-COLING 2024): adaptive truncation of seen vs unseen APIs and hierarchy-aware concentration/diversity. Pipeline: instruction → SBERT retrieve top-k APIs → (optional ToolRerank truncate/rerank) → LLM DFSDT/ReAct planner

Gap. SOTA retrievers are turn-amnesic. They do not maintain a success-conditioned co-activation graph or ingest pruned memory from later stages.

Novelty. Session co-activation cache as a first-class prior over tool pairs. Convex fusion of semantic and session scores; λ may grow with session length. ToolRerank-style seen/unseen truncation kept, then applied after session scoring. FCNP memory mixed into the next SATR prior so retrieval is closed-loop.

O1 comparison — SOTA retrieval vs SATR (no NDCG drawn)

Top: Qin ToolLLM + Zheng ToolRerank (turn-amnesic). Bottom: session fusion + co-activation.



Repository: github.com/joyjeni/session-aware-toolbench-rerank

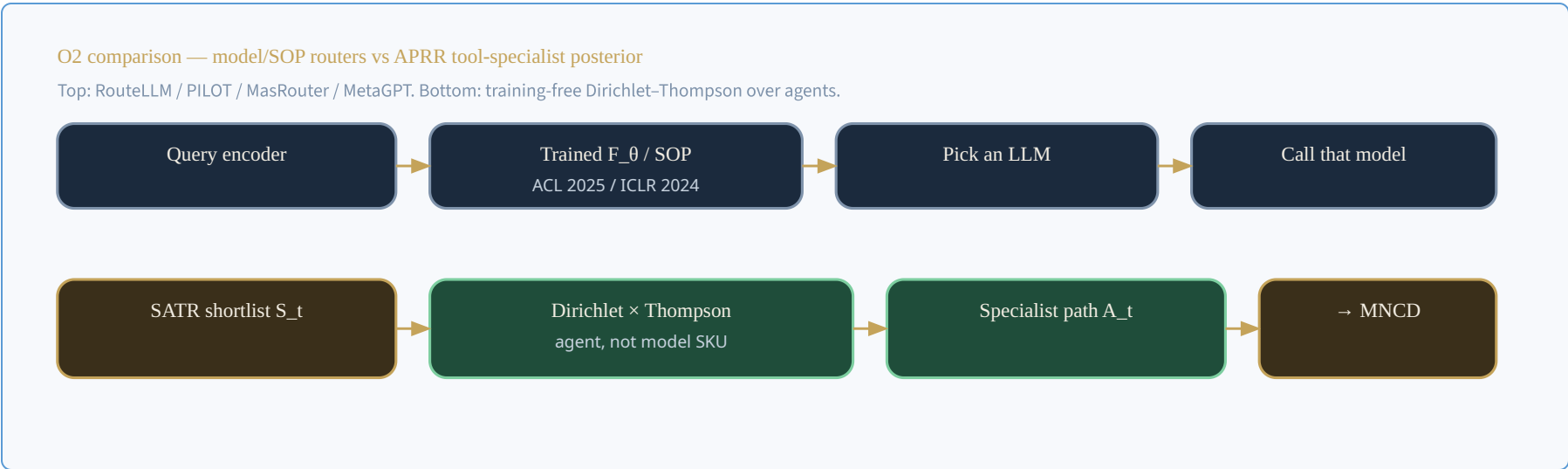
Objective 2 — APRR

The routing chapter: a training-free posterior over tool-specialist agents, updated from SATR scores and from live MNCD observations. The unit of publication is the Bayesian controller, not a Pareto chart.

SOTA. Yue et al., MasRouter (ACL 2025, doi:10.18653/v1/2025.acl-long.757): trained neural controller over multi-agent topologies. Ong et al., RouteLLM (ICLR 2025): routers among LLMs. Panda et al., Adaptive LLM Routing under Budget Constraints (PILOT; Findings of EMNLP 2025, doi:10.18653/v1/2025.findings-emnlp.1301): preference-prior LinUCB for budget-constrained LLM routing. Hong et al., MetaGPT (ICLR 2024): authored SOP workflows. Pipeline: query → trained controller or difficulty model → choose one LLM/agent → execute

Gap. SOTA either trains a neural router, picks a model, or follows an authored SOP. It does not maintain a training-free affinity matrix over Indian OGD tool families.

Novelty. Training-free online W versus MasRouter's learned controller. Implemented update is $\kappa \cdot \text{reward} \cdot 1/L^2 \cdot 1/\text{lat_norm}$ with negative reward on failure. CTGR and FTDR remain in the GitHub repo; this lab runs core APRR hops. W is session state, so routing can adapt across farmer turns.



Repository: github.com/joyjeni/aprr-multi-agent-routing

Objective 3 — MNCD

The consensus-and-execution chapter: gossip (toolId, score), score-sum consensus, live data.gov.in GET on verified UUIDs. The unit of publication is the protocol plus the citation contract.

SOTA. Wang et al., A survey on large language model based autonomous agents (Frontiers of Computer Science, 2024, doi:10.1007/s11704-024-40231-1): construction of LLM agents. He, Treude and Lo, LLM-Based Multi-Agent Systems for Software Engineering (ACM TOSEM, 2025, doi:10.1145/3712003): LMA systems across the SDLC. Guo et al., Large Language Model based Multi-Agents (IJCAI-24, doi:10.24963/ijcai.2024/890): profiling and communication. Chang and Geng, SagaLLM (PVLDB 2025, doi:10.14778/3750601.3750611): transactional context for multi-agent LLM planning. Wu et al., AutoGen (COLM 2024): multi-agent conversation. Hong et al., MetaGPT (ICLR 2024): SOP pipeline. Qian et al., ChatDev (ACL 2024): organisational chat-chain. Li et al., CAMEL (NeurIPS 2023): communicative role-playing agents. Pipeline: manager LLM → sequential or star-topology agent messages → tool calls

Gap. Star and chat topologies coordinate over natural-language messages. They do not vote over tool identifiers backed by a ministry API.

Novelty. First-class vote object is a tool ID, not a chat utterance. consensus_pick is score-sum; consensus_pick_borda is a non-default variant. Live Indian OGD only. Catalog-only tools stay ranking-only; prices are never invented. HTTP 5xx/429 are retried; empty filters show other live rows from the same resource.

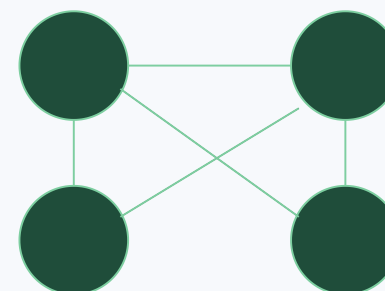
O3 comparison — star/chat SOTA vs mesh vote over live tool IDs

Left: AutoGen / MetaGPT / ChatDev / CAMEL. Right: MNCD gossip + score-sum + data.gov.in.



Vote object = natural-language message

MNCD mesh



Vote object = toolId; execute live OGD UUID

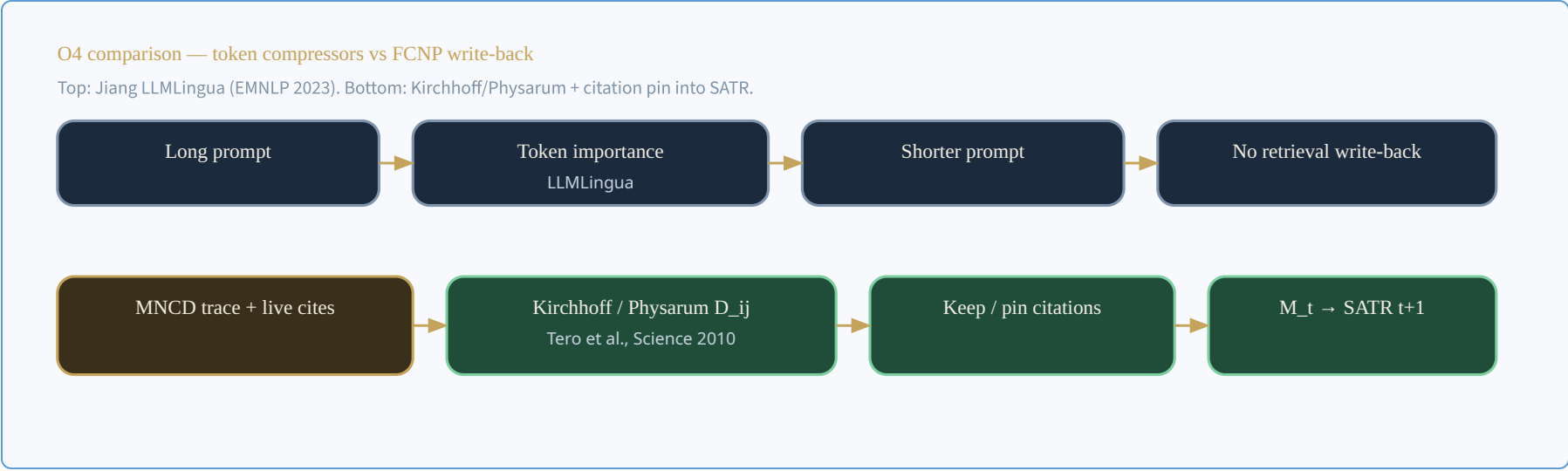
Objective 4 — FCNP

The memory chapter: Kirchhoff/Physarum conductances prune the mesh trace and write surviving live citations back into SATR. The unit of publication is the coupling, not a token-percentage.

SOTA. Jiang et al., LLMingua (EMNLP 2023): token-level prompt compression. Tero et al., Science 2010 (doi:10.1126/science.1177894): Physarum adaptive network. Park et al., Generative Agents (UIST 2023): language memory stream in a sandbox. Pipeline: long prompt → compressor → LLM. Memory is not written back into a tool retriever.

Gap. Token compressors are not current-reinforced over a context graph and do not pin live government citations into the next retrieval turn.

Novelty. Laplacian solve with grounded sink, matching the published Python pruner. Hybrid tiering and persistent high-flow citations. If a requested crop/state has no AGMARKNET rows today, other live rows are shown; nothing is invented. Closed loop: retained spans become SATR session memory.



Repository: github.com/joyjeni/fcnp-context-pruning

Research Questions

Each question is paired with one objective. Answers will be empirical after a protocol is frozen; this slide does not pre-commit scores.

RQ1. Can tool retrieval be conditioned on a co-activation cache and session memory rather than a single query embedding (Qin et al., ToolLLM, ICLR 2024; Zheng et al., ToolRerank, LREC-COLING 2024)? → addressed by O1 (SessionRerank+ (SATR)).

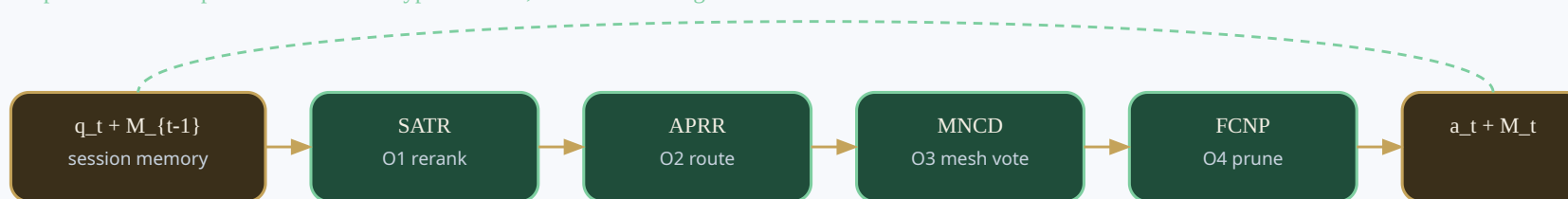
RQ2. Can routing sample a training-free posterior over tool-specialist agents instead of a trained neural controller or an authored SOP (Yue et al., MasRouter, ACL 2025; Hong et al., MetaGPT, ICLR 2024; Ong et al., RouteLLM, ICLR 2025)? → addressed by O2 (Adaptive Probabilistic Routing Reinforcement (APRR)).

RQ3. Can execution proceed as gossiped (toolId, score) plus score-sum consensus, then a live data.gov.in GET, instead of a central chat orchestrator (Wu et al., AutoGen, COLM 2024)? → addressed by O3 (Mesh Network Context Diffusion (MNCD)).

RQ4. Can Kirchhoff/Physarum pruning pin live citations back into retrieval rather than only shortening the prompt (Jiang et al., LLMingua, EMNLP 2023; Tero et al., Science 2010)? → addressed by O4 (Flow-Coupled Network Pruning (FCNP)).

One methodology per objective, then one integrated protocol. Proposal-stage: design the loop and freeze the protocol; do not pre-commit a leaderboard.

Proposed ACRS loop — each arrow is a typed artefact, not a chat message



O1 SATR — fused session ranking against a ToolBench-style library. Live mandi rows are not SATR's job.

O2 APRR — training-free Dirichlet–Thompson posterior over named specialists. Learned routers are a future bake-off, not the implementation.

O3 MNCD — gossip + score-sum consensus, then live data.gov.in GET on verified Agriculture UUIDs. Fail loud. No dummy prices.

O4 FCNP — Kirchhoff / Physarum-inspired conductance prune with write-back into SATR. Not LLMingua token deletion.

Integrated contract. SATR → APRR → MNCD → FCNP must run in that order on one user turn. Skipping MNCD or FCNP is an incomplete run.

Implementation. The four modules are the functions in satr.ts, aprr.ts, mncd.ts, fcnp.ts — formulas on the next slides, full traces on /walkthrough.

What will not appear in this proposal. Promised retrieval scores, latency targets, consensus percentages, or token-reduction ratios.

Research Methodology — O1 SATR

How session-aware ranking will be designed and later evaluated.

- Design method. Specify a fused session object: query + dialogue turns + last tool traces + co-activation counts. Rank tools with that object, not with the raw utterance alone.
- Ranking library (not live prices). Use ToolBench / ToolLLM artefacts as a public tool-ranking library and protocol family. RapidAPI-style traces are ranking evidence only.
- Live Indian OGD is out of SATR's ranking path. Mandi and weather rows enter at MNCD. SATR must not invent or cache dummy AGMARKNET prices.
- Implementation path. Persist co-activation in a session store; expose a rank(query, session) API that APRR can call. Fail loud if the session schema is incomplete.
- Future evaluation protocol (not a result). When the protocol is frozen, compare session-fused ranking against query-only ranking on the same ToolBench-style split. Report the protocol, not a pre-committed score.
- Deliverable for the thesis chapter. Algorithm, schema, and ablation plan (with vs without tool-trace fusion).

Research Methodology — O2 APRR

How the specialist posterior will be designed. No win-rate or millisecond target is claimed at proposal stage.

Design method. Maintain a Dirichlet–Thompson posterior over named specialists. Sample or take the MAP specialist given SATR’s fused session.

Allocation rule (proposal form). $P(\text{agent} \mid \text{context}) \propto W^\alpha \cdot \eta^\beta \cdot \psi^\gamma$, with W = session-matched specialist weight, η = reliability, ψ = cost/risk. Exponents are design knobs, not fitted claims.

Training-free stance. Do not train a MasRouter- or RouteLLM-style classifier as the primary method. Learned routers remain a future bake-off class, not the implementation.

What is being routed. Specialists in the ACRS mesh — not LLM SKUs (GPT-4 vs Mixtral) and not AutoGen conversation modes as the object of routing.

Update rule. After MNCD returns verified or failed evidence, update η (and optionally W) so the next turn’s posterior is not identical to a cold start.

Future evaluation protocol (not a result). Log specialist choice vs task type on held-out session traces; compare against a static role graph. No pre-committed accuracy.

Research Methodology — O3 MNCD

How live Indian OGD and score-sum consensus will be executed. PECAD is a domain precedent, not a baseline to beat.

Design method. Specialists selected by APRR gossip partial beliefs. Aggregate with score-sum (not Borda). Every live claim must cite a verified data.gov.in resource UUID.

Corpus / API. Agriculture-only Open Government Data: AGMARKNET (9ef84268-d588-465a-a308-a864a43d0070) and other UUIDs that pass verification.

Fail-loud contract. HTTP 5xx/429 are retried with backoff; missing API key, unverified UUID, or empty filtered universe hard-fail. No dummy mandi prices.

Platform constraint. data.gov.in Elastic max_result_window is 10 000; limit is capped. Pagination is sequential, not a fabricated parallel harvest.

Related work used correctly. Guo, Woodruff & Yadav, PECAD (AAAI 2020) shows AGMARKNET as a real DSS input. MNCD does not re-implement crop-yield prediction.

Future evaluation protocol (not a result). Measure citation completeness, fail-loud rate on injected faults, and qualitative agreement of score-sum vs majority vote.

Research Methodology — O4 FCNP

How mesh pruning will be designed. No token-reduction ratio is claimed at proposal stage.

Design method. Treat the post-MNCD mesh as a flow network. Apply a Kirchhoff / Physarum-inspired conductance update; drop low-conductance specialist edges; keep a residue.

Write-back (the integration hinge). The residue is written into SATR's next fused session. Without write-back, FCNP would be prompt compression by another name.

Contrast with LLMLingua. LLMLingua shortens tokens before the LLM. FCNP prunes who remains in the mesh. Tokenisers are not the primary artefact.

Heuristic honesty. Discrete conductance is a design heuristic inspired by Tero et al. (Science, 2010), not a proof that the mesh is a Physarum organism.

Safety. Pruning must not delete the last live-data specialist if MNCD still has an open verified query. Fail loud rather than silently drop evidence.

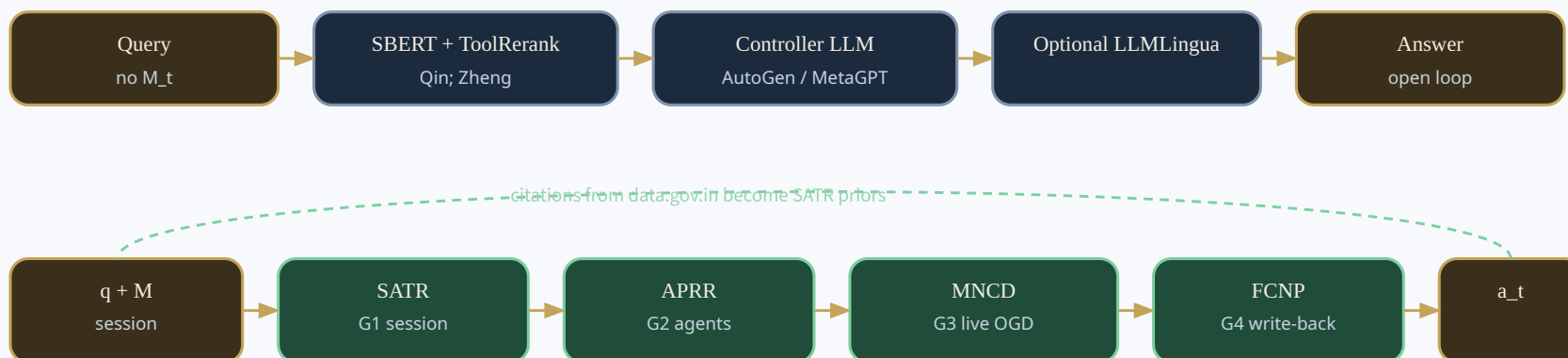
Future evaluation protocol (not a result). Compare mesh size and downstream SATR rank stability with vs without pruning on the same session traces.

Research Methodology — integrated loop

Closed contract SATR → APRR → MNCD → FCNP → SATR. The novelty is the loop and the data contract, not a claimed accuracy.

Integrated contrast — SOTA star/SOP/chat (top) vs ACRS four-objective loop (bottom)

Novelty is the closed contract + live Indian OGD, not a claimed accuracy



Grey: published SOTA. Green: this proposal. No numerical targets are drawn.

Closed-loop protocol. One user turn must traverse all four modules in order. Skipping MNCD (no live UUID) or FCNP (no write-back) is treated as an incomplete run, not a successful demo.

Two evidence regimes. (1) ToolBench-style ranking traces for SATR. (2) Live data.gov.in Agriculture rows for MNCD. Do not mix dummy prices into (1) or RapidAPI tools into (2).

Independent variables (future experiments). Session fusion on/off; Dirichlet routing vs static roles; score-sum vs majority; FCNP write-back on/off.

Dependent measures (to be frozen later). Citation completeness, fail-loud correctness, rank stability across turns, qualitative specialist-choice logs.

Domain lock. Agriculture on Indian OGD for the live path. Other sectors are out of scope until a later amendment.

Ethics / data. Public government catalogues only; no personal data; API keys stay in the environment, never in the thesis text.

Research Methodology – implementation formulas

How the proposal will be implemented: the equations copied from the laboratory, not a promised leaderboard. Constants are the repository defaults.

Module	Formula as coded	Constants
O1 SATR	$s(a q,H)=w_base\ s_base + w_cat\ cat + w_sch\ sch + w_ept\ ept + w_cooc\ \Sigma\ \gamma^{\{n-i\}} \log(1+w_{\{h_i,a\}}) + w_rec\ rec - 0.35\ fails; s_base=0.7\ BM25+0.3\ TFIDF-cos+0.08\ mem-cos$	$w=(1, 0.45, 0.25, 0.3, 0.35, 0.25); \gamma=0.7; \rho=0.02; \delta=1; decay=0.85; BM25\ k1=1.5\ b=0.75$
O2 APRR	$P(a_j a_i,q) \propto W_{ij}^{\alpha} \eta_{ij}^{\beta} \psi_j(q)^{\gamma}; W \leftarrow (1-\lambda)W + \kappa \cdot reward \cdot 1/L^2 \cdot 1/lat$	$\alpha=2, \beta=1, \gamma=2.5, \lambda=0.005, \kappa=5, W0=0.1, \epsilon=0\ (lab), maxHops=4; reward\ +1\ /\ -0.05$
O3 MNCD	$s=0.45\ score/(score +2)+0.35\ overlap+liveBoost-0.05\ idx; tally=\Sigma\ w_a\ s_a; w=success/(1+lat/1000)$	$liveBoost\ 0.25+0.20\ preferred; fanout=3; R=2; \tau=0.55; score-sum\ not\ Borda; live\ UUID\ only$
O4 FCNP	$D_{ij}(t+1)=(1-\mu)D_{ij}+\alpha Q_{ij} ^{\gamma}; L\ p = l; Q= D(p_i-p_j) ; keep\ 35\% / summarize\ 20\% / drop$	$\mu=0.1, \alpha=0.5, \gamma=1.2, sim \geq 0.12; pinned\ live\ citations\ never\ evicted; memory \rightarrow SATR$

Source: lib/research/satr.ts, aprr.ts, mncd.ts, fcnp.ts. Full walkthrough: /walkthrough.

Research Methodology — ToolBench walkthrough

One ToolBench-schema datum, processed by all four objectives. RapidAPI is never GET. Numbers are a 07 September 2026 lab trace, not a metric claim.

Intake. Bundled G1 jsonl qid=6491 is a RapidAPI aircraft query (gold docs 4308–4317). Off-sector flights are stripped. The ranking-library analogue walked here is tb.agri.soil_health with q = “What is the soil pH and recommended fertilizer dose for a farm village?”

O1 SATR. Cold start so $s=z(s_base)$. tb.agri.soil_health $s_base=18.08 \rightarrow s=5.40$ (rank 1, ranking-only). karnataka::shc_karnataka $s=1.40$. datagov.fertilizer $s=0.60$ (live, rank 4).

O2 APRR. Path agriculture_analyst $\rightarrow$ schema_planner ($p=0.73$) $\rightarrow$ retrieval_specialist ($p=0.95$). Non-Agriculture specialists fall back to SATR #1 = soil_health.

O3 MNCD. Agent score $s=0.45$ score/(|score|+2)+0.35 overlap. soil_health 0.5995; tally= $3 \cdot 0.7407 \cdot 0.5995=1.332$. Not liveExecutable. preferredLiveToolId matches fertilizer $\rightarrow$ GET UUID 2e0e6c04-97f2-456b-9309-bf605650cb11 (44 live subsidy rows, e.g. 2002-03 Indigenous Urea 7790 Rs crore).

O4 FCNP. 10 spans $\rightarrow$ keep 6 / drop 4; pin query + live fertilizer observation + citation. Memory written back to SATR.

ToolBench = ranking library. Live evidence is always a verified data.gov.in Agriculture UUID.

Research Methodology — data.gov.in walkthrough

The same four objectives on live AGMARKNET. Query: “What is the current mandi price of wheat in Punjab?” Fail-loud: no invented Punjab-wheat modal.

Intake. extractToolArguments → state=Punjab, commodity=Wheat. preferredLiveToolId → datagov.mandi_prices (UUID 9ef84268-d588-465a-a308-a864a43d0070). Limit capped at 10 000.

O1 SATR. karnataka::agmarknet_ka s=3.73 (rank 1, same UUID); datagov.mandi_prices s=3.05 (rank 2, preferred); datagov.msp s=2.31 (ranking-only).

O2 APRR. Path agriculture_analyst → schema_planner (p=0.54) → tool_executor (p=0.66, terminal stop). Hop 0: Karnataka mandi, national mandi, MSP. Hop 2 live leftover: crop_production.

O3 MNCD. mandi_prices agent-score 0.787 (liveBoost 0.45); karnataka 0.7789. Score-sum winner karnataka::agmarknet_ka tally=1.154. Live GET: 10 000 arrivals; 0 Wheat in Punjab today; 479 other live Punjab rows; Wheat in 133 live rows from MP, Rajasthan, UP, Gujarat, Maharashtra, West Bengal, Chhattisgarh. Shown mean modal Rs 2593/quintal (e.g. Bhindi, Dera Baba Nanak APMC, Rs 828, 07/09/2026).

O4 FCNP. 10 spans → keep 7; pin the AGMARKNET citation. $W \leftarrow (1-\lambda)W + \kappa \cdot 1/L^2 / \text{lat}$ on the hop path. Next SATR is session-conditioned.

Lab trace 07 September 2026. Re-run on /walkthrough; rows change daily. No dummy prices.

Conclusion

ACRS is proposed as a structural orchestration layer. The contribution is the closed loop and the four named gaps — not a pre-committed leaderboard.

The literature (Wang et al. FCS 2024; He et al. TOSEM 2025; Guo et al. IJCAI-24; SagaLLM PVLDB 2025; AutoGen, MetaGPT, ChatDev, CAMEL, ToolLLM, MasRouter, RouteLLM, PILOT, LLMLingua, PECAD) establishes agent construction, conversation, tools, learned routing, and prompt compression. It does not establish SATR → APRR → MNCD → FCNP as one contract over live Indian OGD.

The identified problem is five gaps: G1 session-tool fusion, G2 training-free specialist posterior, G3 live fail-loud Indian OGD, G4 conductance prune with write-back, G5 missing integration layer.

The aim is to design that layer. Four objectives (SATR, APRR, MNCD, FCNP) and four research questions map onto it.

Methodology is specified per objective: ToolBench as ranking library; Dirichlet-Thompson routing; verified data.gov.in UUIDs with score-sum; Physarum-inspired prune with SATR write-back. Metric numbers are deferred.

Next step after approval. Freeze evaluation protocols, implement the closed loop, and report whatever the measurements show — including negative results.

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CrewAI is an engineering framework without a flagship peer-reviewed paper in this list; AutoGen and MetaGPT are the MAS citations.

Thank you

Adaptive Context Reasoning System — a research programme, not a performance number.

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